

A Preventive Maintenance Model with Periodic and Random Inspection Policy for a Three-stage Failure Process

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Abstract—Inspection is an effective method to guarantee the safety and reliability of the components and systems. For some systems executing jobs with random working times, implementing a comprehensive inspection at fixed periodic interval would disrupt the ongoing job. Therefore it would be cost-effective to inspect these systems thoroughly at the completion of every N jobs, which is named as random inspection policy. Considering such systems, a new model is proposed by combining the random and periodic inspection policy based on a three-stage failure process. The whole life cycle of the system is divided into three independent phases: normal, minor and severe defective stages. Two-level inspections are employed in this model. The minor inspections are performed at a fixed periodic interval, which can identify the minor defective stage with a certain probability but detect the severe defective stage perfectly. The major inspections are performed when every N jobs is completed, which can completely reveal both minor and severe defective stages. When the system is found in minor defective stage, the minor inspection interval will be shortened. If the severe defective stage is identified, instead of maintaining the system immediately, the maintenance action will be postponed to the completion of the current job. Numerical examples and comparisons with other models are presented to demonstrate the correctness, generality and economy of the model.

Keywords—random inspection; three-stage failure process; postponed maintenance; random working time

I. INTRODUCTION

To improve the reliability and economic performance of the components and systems, preventive maintenance (PM) policies are commonly adopted in industry. Many PM models about repair, replacement, and inspection were summarized [1, 2]. Inspection has always been playing an important role in PM models, which provides information about system state to arrange appropriate maintenance action. An effective inspection scheme implemented at certain time can actually prevent unexpected failure as well as avoid excessive maintenance.

There are many PM models consider inspections with different depths or levels at fixed intervals [3-6]. However, for some systems successively executing jobs with random working times, inspecting the whole system comprehensively at a strict fixed interval is costlier or impractical. Such systems are quite common in industry, for example, a machine tool

processing workpieces one by one, a plane conducting a flight mission or a train being on a journey again and again. The job executed in such systems has a random working cycle or processing time, implementing a comprehensive inspection at fixed periodic interval would disrupt the ongoing job. It is therefore rational to consider that inspection happening at the completion of every N jobs is more economical and practical for such systems. This inspection is named as random inspection [7].

Different from the binary state (normal or failed) system, the degradation process of some systems is usually divided into more than two stages [8, 9]. This degradation process is named as delay time stage, which was early applied to PM model in [10]. Christer [11] defined the normal stage and delay time stage, and used this concept to predict an efficient inspection frequency. Wang [12, 13] reviewed the recent advances in delay-time-based PM models, and extended the two-stage delay time concept to a three-stage failure process to reflect some actual maintenance scenarios. A three-stage failure process means the failure process can be divided into three independent stages, namely normal, minor and severe defective stages. The normal stage is defined as the time from new to the initial point that minor defect can be identified, and the time from that point to the point that system turns to be severe defect is named as minor defective stage, then the severe defective stage starts and lasts until the system fails. In practice, the separation of different stages could base on the threshold of noise or vibration of the system. Furthermore, for a redundant system, the initial point of a redundant subsystem failure could be regarded as a critical point distinguishing adjacent stages. Then the concept of imperfect inspection and planned PM are introduced to the PM model based on a three-stage failure process [14].

The merits of postponed maintenance are summarized [15]: to avoid production disruption or unnecessary or ineffective replacement; to prepare for replacement; to extend component life; to wait for an opportunity. In order not to interfere with ongoing task, the maintenance action is allowed to postpone to the completion of the current job if necessary.

Two-level inspection policy is considered in this paper, where a minor inspection happens at a fixed interval and a major inspection is implemented at the completion of every N jobs. The minor inspection could be a simple online inspection, which is low-cost and don't interfere with ongoing job. Minor inspection is supposed to identify the minor defective stage

with a limited probability but can always distinguish a severe defective stage perfectly. Major inspection is a comprehensive detection, which can completely identify both the minor and the severe defective stages. Once the system is found in the minor defective stage by either inspection, the minor inspection interval will be shortened, which provides more chances to identify the severe defective stage. When the severe defective stage is identified, the maintenance action will not be taken until the present job is completed, because implementing maintenance action immediately may be costly and impractical for these systems executing jobs with random working times.

The new contributions of this paper can be summarized as:

- The random inspection policy is introduced to this model based on three-stage failure process; thereby the application of this mode is extended to the systems executing jobs with random working times.
- The model proposed in this paper is more general. When set working time of job a particular value, our model is also applicable to the system that does not consider random working times. Therefore the situation where the random working times of jobs are not considered can be regard as a special case of our model.
- The major inspection is arranged at the completion of every N jobs, and its cost may be lower compared with implementing major inspection at fixed periodic interval.
- When the system is found in the severe defective stage, instead of implementing a maintenance action immediately, the system will be renewed when the current job is completed, which is named as postponed maintenance in this paper.

This paper is organized as follows. Section 2 offers the modeling assumptions and notations. The periodic and random inspection model based on a three-stage failure process is described in section 3. Section 4 presents numerical examples to demonstrate the correctness and indicate the generality and economy of this model. Finally, a conclusion and the key points of further research are given in section 5.

II. MODEL ASSUMPTIONS AND NOTATIONS

For a model building purpose, some assumptions and notations are proposed. Most assumptions are consistent with the situation in the practical industry and have been applied in previous researches [7, 14].

- The whole life cycle of the system is divided into three independent phases: normal, minor and severe defective stages.
- The system is checked at fixed interval T by minor inspection. The minor inspection can be performed online and doesn't affect the system jobs, so the cost of minor inspection is relatively low. The system is checked by major inspections when every N jobs is completed. Downtime cost should also be considered as part of the cost of major inspection. For simplicity, the downtime doesn't appear on timeline. When

major inspection and minor inspection happen at the same time, only the major inspection is implemented.

- Minor inspection can detect minor defective stage with a certain detection probability γ , and β is the probability of finding no fault ($\beta = 1 - \gamma$), while major inspection is supposed to detect minor and severe defective stages perfectly.
- When the system is found in minor defective stage, the minor inspection interval will be shortened, for simplicity, the interval turns $T/2$ in this paper. If the minor defective stage is identified by a minor inspection, its interval becomes shorter immediately, otherwise the interval will be shortened at the next minor inspection.
- If the minor or the major inspection finds the system in severe defective stage, the system will be renewed when the executing job is finished.
- The failure of the system could be identified immediately, and the only maintenance action is replacement. The cost of replacing the failed system is significantly higher than that of replacing the defective system.
- The system executes jobs with random times $Y_j, j = (1, 2, \dots)$. Y_j has an independent and identical distribution $F(t) = P(Y_j \leq t)$ with a finite mean $1/\theta$. $S_n = \sum_{i=1}^n Y_i$ denotes the sum of $Y_1 + Y_2 + \dots + Y_n$, and its distribution can be expressed by $F^{(n)}(t)$. $F^{(n)}(t)$ is the n -fold Stieltjes convolution of $F(t)$ with itself, that is $F^{(n)}(t) = \int_0^t F^{(n-1)}(t-u)dF(u)$, $n = (1, 2, \dots)$, where $F^{(0)}(t) = 1 (t > 0)$. The probability that the system completes n jobs in the interval $[0, t)$ can be expressed as $F^{(n)}(t) - F^{(n+1)}(t)$.

The notations used in this paper are shown in Table I.

TABLE I. THE NOTATIONS USED IN THIS PAPER

X_n	random variable representing the duration of the n th stage of the system
$f_{X_n}(x)$	probability density function (pdf) of X_n
T	interval of minor inspection
Y_j	random time of a job
S_n	the sum of $Y_1, Y_2, \dots, Y_n$
$F^{(n)}(t)$	distribution of S_n
γ	probability of minor inspection detecting minor defective stage
T_f	random time that the failed system is renewed
T_r	random time that the defective system is renewed
c_{mi}	cost of a minor inspection
c_{ma}	cost of a major inspection
c_f	cost of the failed system being renewed at T_f
c_r	cost of the defective system being renewed at T_r

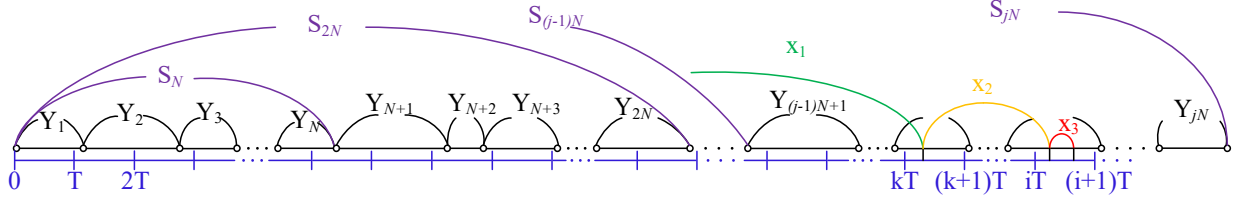


Figure 1. The system fails at T_f , $T_f \in (iT, (i+1)T)$.

III. A PM MODEL WITH PERIODIC AND RANDOM INSPECTION POLICY FOR A THREE-STAGE FAILURE PROCESS

In this section, a periodic and random inspection policy based on a three-stage failure process is presented. The inspections contain minor inspection at periodic interval T and major inspection at the completion of every N jobs. The long run expected cost per unit time is selected as a measure to optimize T and N . As described above, the system will be renewed when it fails or it is identified in the severe defective stage; accordingly, the probabilities of these renewal situations are formulated. Then the expected renewal cycle cost and the expected renewal cycle length can be expressed, thereby the long run expected cost per unit time can be derived.

A. Probability of Failure Renewal

According to the time when the minor or severe defective stage is identified by minor or major inspection, the probabilities of system failure are formulated in different situations.

1) The system fails at T_f , $T_f \in (iT, (i+1)T)$. As Fig. 1 shows, before the system fails, neither the minor nor the severe defective stage is detected by any inspection. The normal stage continues until the minor defective stage starts in $[kT, (k+1)T]$. Then the inspections don't detect the minor defective stage until the severe defective stage starts in $[iT, (i+1)T]$. The system fails before the next inspection is executed.

$$P(iT < T_f < (i+1)T)$$

$$= \sum_{k=0}^i P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, \\ iT - X_1 < X_2 < (i+1)T - X_1, \\ i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ 0 < X_3 < (i+1)T - X_1 - X_2, \\ S_{(j-1)N} < X_1, \\ X_1 + X_2 + X_3 < S_{jN} \end{array} \right\} \quad (1)$$

$$= \sum_{k=0}^i \int_{kT}^{(k+1)T} \int_{\Delta(iT-x)}^{(i+1)T-x} \int_0^{(i+1)T-x-y} \beta^{i-k} f_{X_1}(x) \times f_{X_2}(y) f_{X_3}(z) F^{((j-1)N)}(x) F^{(jN)}(x+y+z) dx dy dz$$

where $F^{(jN)}(x+y+z) = 1 - F^{(jN)}(x+y+z)$, because the lower limit of integration must be no less than 0, i.e. $(iT - x) \geq 0$, thus $\Delta(g(x))$ is defined as

$$\Delta(g(x)) = \begin{cases} 0, & g(x) \leq 0 \\ g(x), & g(x) > 0 \end{cases} \quad (2)$$

2) The system fails at T_f , $T_f \in ((i+1)T, S_{(j-1)N+n})$. The inspections don't detect the minor defective stage until the severe defective stage starts in $[iT, (i+1)T]$. Then the severe defective stage is detected by minor inspection first, but it is decided to stop minor inspection and renew the system until the current job is completed. Unfortunately, the system fails before the current job is completed.

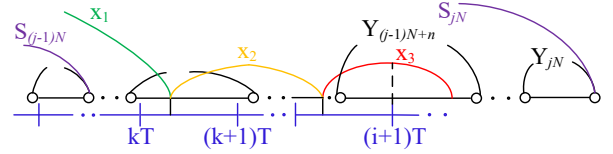
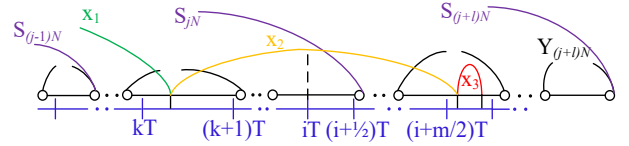


Figure 2. The system fails at T_f , $T_f \in ((i+1)T, S_{(j-1)N+n})$

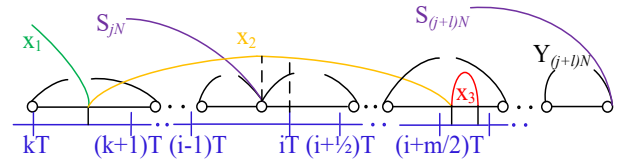
$$P((i+1)T < T_f < S_{(j-1)N+n}) = \sum_{k=0}^i P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, \\ iT - X_1 < X_2 < (i+1)T - X_1, \\ i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ (i+1)T - X_1 - X_2 < X_3 < S_{(j-1)N+n} - X_1 - X_2, \\ S_{(j-1)N} < X_1, \\ S_{(j-1)N+n-1} < (i+1)T \end{array} \right\} \quad (3)$$

$$= F^{((j-1)N+n-1)}((i+1)T) \sum_{k=0}^i \int_{kT}^{(k+1)T} \int_{\Delta(iT-x)}^{(i+1)T-x} \int_{\Delta((i+1)T-x-y)}^{S_{(j-1)N+n}-x-y} \beta^{i-k} \times f_{X_1}(x) f_{X_2}(y) f_{X_3}(z) F^{((j-1)N)}(x) dx dy dz$$

3) The system fails at T_f , $T_f \in ((i+m/2)T, (i+(m+1)/2)T)$. Before the system fails, only the minor defective stage is detected by minor or major inspection.



(a) The minor defective stage is first identified by minor inspection



(b) The minor defective stage is first identified by major inspection

Figure 3. The system fails at T_f , $T_f \in ((i + \frac{m}{2})T, (i + \frac{m+1}{2})T)$

When the minor inspection detects the minor defective stage at iT firstly, the minor inspection interval become $T/2$ immediately. The severe defective stage starts in $[(i + m/2)T, (i + (m + 1)/2)T]$, and the system fails before the next inspection is executed.

$$P_1((i + \frac{m}{2})T < T_f < (i + \frac{m+1}{2})T) \\ = \sum_{l=0}^{\infty} \sum_{k=0}^{i-1} P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, \\ (i + \frac{m}{2})T - X_1 < X_2 < (i + \frac{m+1}{2})T - X_1, \\ i - k \text{ minor inspetions don't detect } X_2 \text{ stage,} \\ 0 < X_3 < (i + \frac{m+1}{2})T - X_1 - X_2, \\ S_{(j-1)N} < X_1, iT < S_{jN}, \\ X_1 + X_2 + X_3 < S_{(j+l)N}, S_{(j+l-1)N} < X_1 + X_2, \end{array} \right\} \\ = \sum_{l=1}^{\infty} \sum_{k=0}^{i-1} \int_{kT}^{(k+1)T} \int_{\Delta((i+\frac{m}{2})T-x)}^{(i+\frac{m+1}{2})T-x} \int_0^{(i+\frac{m+1}{2})T-x-y} \beta^{i-k} f_{X_1}(x) f_{X_2}(y) f_{X_3}(z) \times \\ F^{((j-1)N)}(x) F^{(jN)}(iT) F^{((j+l-1)N)}(x+y) F^{((j+l)N)}(x+y+z) dx dy dz \quad (4)$$

When the minor defective stage is first detected by major inspection at S_{jN} , the minor inspection interval will become $T/2$ at the next minor inspection.

$$P_2((i + \frac{m}{2})T < T_f < (i + \frac{m+1}{2})T) \\ = \sum_{l=0}^{\infty} \sum_{k=0}^{i-1} P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, S_{(j-1)N} < X_1 < S_{jN}, \\ (i + \frac{m}{2})T - X_1 < X_2 < (i + \frac{m+1}{2})T - X_1, \\ i - k \text{ minor inspetions don't detect } X_2 \text{ stage,} \\ 0 < X_3 < (i + \frac{m+1}{2})T - X_1 - X_2, \\ (i-1)T < S_{jN} < iT, \\ X_1 + X_2 + X_3 < S_{(j+l)N}, S_{(j+l-1)N} < X_1 + X_2 \end{array} \right\} \\ = \sum_{l=1}^{\infty} \sum_{k=0}^{i-1} [F^{(jN)}(iT) + F^{(jN)}((i-1)T) - 1] \times \\ \int_{kT}^{(k+1)T} \int_{\Delta((i+\frac{m}{2})T-x)}^{(i+\frac{m+1}{2})T-x} \int_0^{(i+\frac{m+1}{2})T-x-y} \beta^{i-k} f_{X_1}(x) f_{X_2}(y) f_{X_3}(z) \times \\ F^{((j-1)N)}(x) F^{(jN)}(x) F^{((j+l-1)N)}(x+y) F^{((j+l)N)}(x+y+z) dx dy dz$$

The probability that system fails at $T_f \in ((i + m/2)T, (i + (m + 1)/2)T)$ is:

$$P((i + \frac{m}{2})T < T_f < (i + \frac{m+1}{2})T) \\ = P_1((i + \frac{m}{2})T < T_f < (i + \frac{m+1}{2})T) + P_2((i + \frac{m}{2})T < T_f < (i + \frac{m+1}{2})T) \quad (6)$$

4) The system fails at T_f , $T_f \in ((i + (m + 1)/2)T, S_{(j+l-1)N+n})$. Before the system fails, both the minor and severe defective stage are detected. The minor or major inspection detects the minor defective stage in $[(i - 1)T, iT]$, and the severe defective stage is detected by minor inspection at $(i + (m + 1)/2)T$. Then the maintenance is arranged at the

completion of the current job, but the system fails before the job is completed.

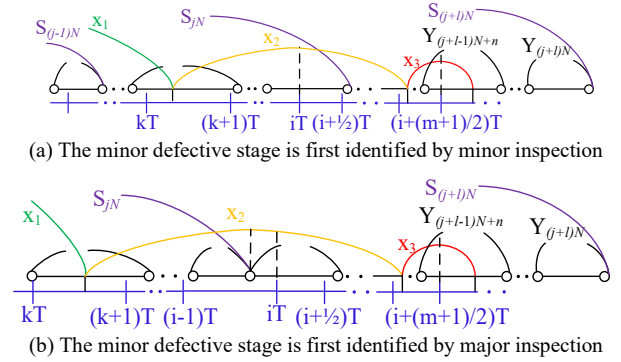


Figure 4. The system fails at T_f , $T_f \in ((i + \frac{m+1}{2})T, S_{(j+l-1)N+n})$

The minor defective stage is first detected by minor inspection:

$$P_1((i + \frac{m+1}{2})T < T_f < S_{(j+l-1)N+n}) \\ = \sum_{k=0}^{i-1} P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, S_{(j-1)N} < X_1, \\ (i + \frac{m}{2})T - X_1 < X_2 < (i + \frac{m+1}{2})T - X_1, \\ i - k \text{ minor inspetions don't detect } X_2 \text{ stage,} \\ (i + \frac{m+1}{2})T - X_1 - X_2 < X_3 < S_{(j+l-1)N+n} - X_1 - X_2, \\ iT < S_{jN}, S_{(j+l-1)N} < X_1 + X_2, \\ S_{(j+l-1)N+n-1} < (i + \frac{m+1}{2})T \end{array} \right\} \\ = F^{(jN)}(iT) F^{((j+l-1)N+n-1)}((i + \frac{m+1}{2})T) \times \\ \sum_{k=0}^{i-1} \int_{kT}^{(k+1)T} \int_{\Delta((i+\frac{m}{2})T-x)}^{(i+\frac{m+1}{2})T-x} \int_{\Delta((i+\frac{m}{2})T-x)}^{S_{(j+l-1)N+n}-x-y} \beta^{i-k} \times \\ f_{X_1}(x) f_{X_2}(y) f_{X_3}(z) F^{((j-1)N)}(x) F^{((j+l-1)N)}(x+y) dx dy dz \quad (7)$$

The minor defective stage is first detected by major inspection:

$$P_2((i + \frac{m+1}{2})T < T_f < S_{(j+l-1)N+n}) \\ = \sum_{k=0}^{i-1} P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, S_{(j-1)N} < X_1 < S_{jN}, \\ (i + \frac{m}{2})T - X_1 < X_2 < (i + \frac{m+1}{2})T - X_1, \\ i - k \text{ minor inspetions don't detect } X_2 \text{ stage,} \\ (i + \frac{m+1}{2})T - X_1 - X_2 < X_3 < S_{(j+l-1)N+n} - X_1 - X_2, \\ (i-1)T < S_{jN} < iT, S_{(j+l-1)N} < X_1 + X_2, \\ S_{(j+l-1)N+n-1} < (i + \frac{m+1}{2})T \end{array} \right\} \\ = F^{(jN)}((i-1)T) F^{(jN)}(iT) F^{((j+l-1)N+n-1)}((i + \frac{m+1}{2})T) \times \\ \sum_{k=0}^{i-1} \int_{kT}^{(k+1)T} \int_{\Delta((i+\frac{m}{2})T-x)}^{(i+\frac{m+1}{2})T-x} \int_{\Delta((i+\frac{m}{2})T-x)}^{S_{(j+l-1)N+n}-x-y} \beta^{i-k} f_{X_1}(x) f_{X_2}(y) f_{X_3}(z) \times \\ F^{((j-1)N)}(x) F^{(jN)}(x) F^{((j+l-1)N)}(x+y) dx dy dz \quad (8)$$

The probability that system fails at $T_f \in ((i + (m + 1)/2)T, S_{(j+l-1)N+n})$ is:

$$P((i + \frac{m+1}{2})T < T_f < S_{(j+l-1)N+n}) = P_1((i + \frac{m+1}{2})T < T_f < S_{(j+l-1)N+n}) + P_2((i + \frac{m+1}{2})T < T_f < S_{(j+l-1)N+n}) \quad (9)$$

B. Probability of Defective Renewal before Failure

1) The system is renewed at T_r , $T_r = S_{(j-1)N+n}$. Before the system is renewed, only the severe defective stage is detected by minor inspection at $(i + 1)T$. Then it is decided to stop inspection and renew the system at the completion of the current job, consequently the system is replaced at $S_{(j-1)N+n}$ before it fails.

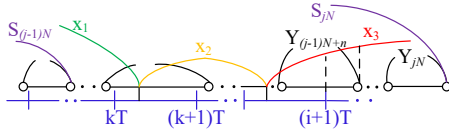


Figure 5. The system is renewed at T_r , $T_r = S_{(j-1)N+n}$

$$P(T_r = S_{(j-1)N+n}) = \sum_{k=0}^i P_1 \left\{ \begin{array}{l} kT < X_1 < (k+1)T, \\ iT - X_1 < X_2 < (i+1)T - X_1, \\ i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ S_{(j-1)N+n} - X_1 - X_2 < X_3, \\ S_{(j-1)N} < X_1, \\ S_{(j-1)N+n-1} < (i+1)T \leq S_{(j-1)N+n} \end{array} \right\} \quad (10)$$

$$= \sum_{k=0}^i [F^{((j-1)N+n-1)}((i+1)T) - F^{((j-1)N+n)}((i+1)T)](\beta)^{i-k} \times \int_{kT}^{(k+1)T} \int_{\Delta((i+1)T-x)}^{\infty} \int_{S_{jN+n}-x-y}^{\infty} f_{X_1}(x)f_{X_2}(y)f_{X_3}(z)F^{(jN)}(x)dx dy dz$$

2) The system is renewed at T_r , $T_r = S_{(j+l-1)N+n}$. Before the system is renewed, both the minor and severe defective stages are detected. The inspection detects the minor defective stage in $[(i-1)T, iT]$, and the severe defective stage is detected by minor inspection at $(i + (m + 1)/2)T$. Instead of maintaining the system immediately, inspection is stopped and the maintenance action is implemented until the current job is completed. Then the system is renewed at $S_{(j+l-1)N+n}$ before it fails.

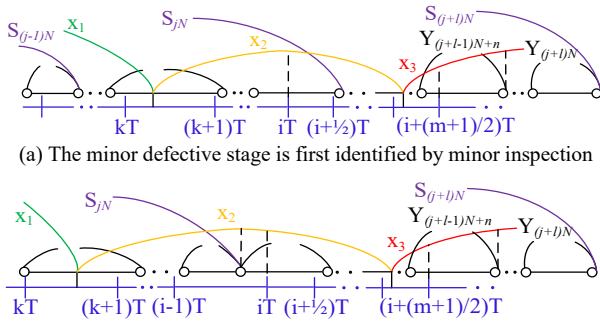


Figure 6. The system is renewed at T_r , $T_r = S_{(j+l-1)N+n}$

The minor defective stage is first detected by minor inspection:

$$P_1(T_r = S_{(j+l-1)N+n}) = \sum_{k=0}^{i-1} P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, \quad S_{(j-1)N} < X_1, \\ (i + \frac{m}{2})T - X_1 < X_2 < (i + \frac{m+1}{2})T - X_1, \\ i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ S_{(j+l-1)N+n} - X_1 - X_2 < X_3, \\ iT < S_{jN}, \quad S_{(j+l-1)N} < X_1 + X_2, \\ S_{(j+l-1)N+n-1} < (i + \frac{m+1}{2})T < S_{(j+l-1)N+n} \end{array} \right\}$$

$$= [F^{((j+l-1)N+n-1)}((i + \frac{m+1}{2})T) - F^{((j+l-1)N+n)}((i + \frac{m+1}{2})T)] \times \quad (11)$$

$$\frac{F^{(jN)}}{F^{(jN)}}(iT) \sum_{k=0}^{i-1} \int_{kT}^{(k+1)T} \int_{\Delta((i+\frac{m}{2})T-x)}^{(i+\frac{m+1}{2})T-x} \int_{\Delta((i+\frac{m+1}{2})T-x-y)}^{\infty} \beta^{i-k} \times f_{X_1}(x)f_{X_2}(y)f_{X_3}(z)F^{((j-1)N)}(x)F^{((j+l-1)N)}(x+y)dx dy dz$$

The minor defective stage is first detected by major inspection:

$$P_2(T_r = S_{(j+l-1)N+n}) = \sum_{k=0}^{i-1} P \left\{ \begin{array}{l} kT < X_1 < (k+1)T, \quad S_{(j-1)N} < X_1 < S_{jN}, \\ (i + \frac{m}{2})T - X_1 < X_2 < (i + \frac{m+1}{2})T - X_1, \\ i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ S_{(j+l-1)N+n} - X_1 - X_2 < X_3, \\ (i-1)T < S_{jN} < iT, \quad S_{(j+l-1)N} < X_1 + X_2, \\ S_{(j+l-1)N+n-1} < (i + \frac{m+1}{2})T < S_{(j+l-1)N+n} \end{array} \right\}$$

$$= [F^{((j+l-1)N+n-1)}((i + \frac{m+1}{2})T) - F^{((j+l-1)N+n)}((i + \frac{m+1}{2})T)] \times \quad (12)$$

$$\frac{F^{(jN)}}{F^{(jN)}}(iT)F^{(jN)}((i-1)T) \sum_{k=0}^{i-1} \int_{kT}^{(k+1)T} \int_{\Delta((i+\frac{m}{2})T-x)}^{(i+\frac{m+1}{2})T-x} \int_{\Delta((i+\frac{m+1}{2})T-x-y)}^{\infty} \beta^{i-k} \times f_{X_1}(x)f_{X_2}(y)f_{X_3}(z)F^{((j-1)N)}(x)F^{(jN)}(x)F^{((j+l-1)N)}(x+y)dx dy dz$$

The probability of the system being renewed at $T_r = S_{(j+l-1)N+n}$:

$$P(T_r = S_{(j+l-1)N+n}) = P_1(T_r = S_{(j+l-1)N+n}) + P_2(T_r = S_{(j+l-1)N+n}) \quad (13)$$

3) The system is renewed at T_r , $T_r = S_{jN}$. Before the system is renewed, the minor defective stage is not detected by any inspection. Then the severe defective stage is first identified by major inspection at S_{jN} and replacement is implemented immediately.

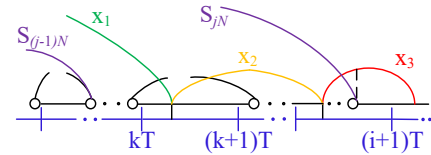
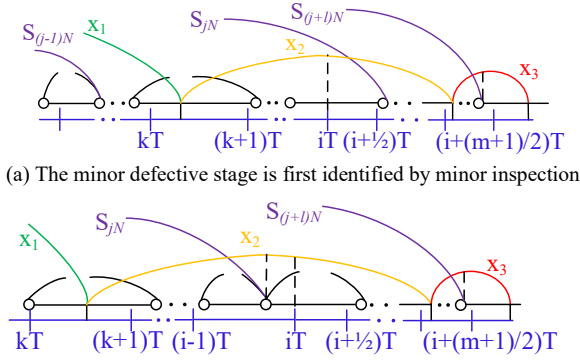


Figure 7. The system is renewed at T_r , $T_r = S_{jN}$

$$\begin{aligned}
P(T_r = S_{jN}) &= \sum_{k=0}^i P_3 \left\{ \begin{aligned} &kT < X_1 < (k+1)T, \\ &iT - X_1 < X_2 < S_{jN} - X_1, \\ &i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ &S_{jN} - X_1 - X_2 < X_3, \\ &S_{(j-1)N} < X_1, \\ &S_{jN} < (i+1)T \end{aligned} \right\} \quad (14) \\
&= F^{(jN)}((i+1)T) \sum_{k=0}^i \int_{kT}^{(k+1)T} \int_{\Delta((i+1)T-x)}^{S_{jN}-x} \int_{S_{jN}-x-y}^{\infty} \beta^{i-k} \times \\
&\quad f_{X_1}(x) f_{X_2}(y) F^{((j-1)N)}(x) dx dy dz
\end{aligned}$$

4) The system is renewed at T_r , $T_r = S_{(j+l)N}$. Before the system is renewed, the minor defective stage is detected by minor or major inspection in $[(i-1)T, iT]$. Then the major inspection first detects severe defective stage at $S_{(j+l)N}$ and the system is renewed immediately.



(a) The minor defective stage is first identified by minor inspection

(b) The minor defective stage is first identified by major inspection

Figure 8. The system is renewed at T_r , $T_r = S_{(j+l)N}$

The minor defective stage is first detected by major inspection:

$$\begin{aligned}
P_1(T_r = S_{(j+l)N}) &= \sum_{k=0}^{i-1} P \left\{ \begin{aligned} &kT < X_1 < (k+1)T, \quad S_{(j-1)N} < X_1, \\ &(i + \frac{m}{2})T - X_1 < X_2 < S_{(j+l)N} - X_1, \\ &i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ &S_{(j+l)N} - X_1 - X_2 < X_3, \\ &iT < S_{jN}, \quad S_{(j+l-1)N} < X_1 + X_2, \\ &S_{(j+l)N} < (i + \frac{m+1}{2})T \end{aligned} \right\} \quad (15) \\
&= F^{(jN)}(iT) F^{((j+l)N)}((i + \frac{m+1}{2})T) \sum_{k=0}^{i-1} \int_{kT}^{(k+1)T} \int_{\Delta((i + \frac{m}{2})T-x)}^{S_{(j+l)N}-x} \int_{S_{(j+l)N}-x-y}^{\infty} \beta^{i-k} \times \\
&\quad f_{X_1}(x) f_{X_2}(y) f_{X_3}(z) F^{((j-1)N)}(x) F^{((j+l-1)N)}(x+y) dx dy dz
\end{aligned}$$

The minor defective stage is first detected by major inspection:

$$\begin{aligned}
P_2(T_r = S_{(j+l)N}) &= \sum_{k=0}^{i-1} P \left\{ \begin{aligned} &kT < X_1 < (k+1)T, \quad S_{(j-1)N} < X_1 < S_{jN}, \\ &(i + \frac{m}{2})T - X_1 < X_2 < S_{(j+l)N} - X_1, \\ &i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ &S_{(j+l)N} - X_1 - X_2 < X_3, \\ &(i-1)T < S_{jN} < iT, \quad S_{(j+l-1)N} < X_1 + X_2, \\ &S_{(j+l)N} < (i + \frac{m+1}{2})T \end{aligned} \right\} \quad (16) \\
&= F^{(jN)}((i-1)T) F^{(jN)}(iT) F^{((j+l)N)}((i + \frac{m+1}{2})T) \times \\
&\quad \sum_{k=0}^{i-1} \int_{kT}^{(k+1)T} \int_{\Delta((i + \frac{m}{2})T-x)}^{S_{(j+l)N}-x} \int_{S_{(j+l)N}-x-y}^{\infty} (\beta)^{i-k} f_{X_1}(x) f_{X_2}(y) f_{X_3}(z) \times \\
&\quad F^{((j-1)N)}(x) F^{(jN)}(x) F^{((j+l-1)N)}(x+y) dx dy dz
\end{aligned}$$

The probability that the system renewed at $T_r = S_{(j+l)N}$:

$$\begin{aligned}
P(T_r = S_{(j+l)N}) &= P_1(T_r = S_{(j+l)N}) + P_2(T_r = S_{(j+l)N}) \quad (17)
\end{aligned}$$

C. The Long Run Expected Cost per Unit Time

As presented above, the pdfs that the system renewed in different situations have been formulated. On this foundation, it is convenient to express the expected renewal cycle cost and the expected renewal cycle length. Accordingly, the long run expected cost per unit time is derived.

The expected cycle cost caused by a failure renewal:

$$\begin{aligned}
E(C_f) &= \left\{ \begin{aligned} &\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} [ic_{mi} + (j-1)c_{ma} + c_f] P(iT < T_f < (i+1)T) + \\ &\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^N [(i+1)c_{mi} + (j-1)c_{ma} + c_f] \times \\ &\quad P((i+1)T < T_f < S_{(j-1)N+n}) + \\ &\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \sum_{m=1}^{\infty} [(i+m)c_{mi} + (j-1)c_{ma} + c_f] \times \\ &\quad P((i + \frac{m}{2})T < T_f < (i + \frac{m+1}{2})T) + \\ &\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^N \sum_{m=1}^{\infty} [(i+m+1)c_{mi} + (j-1)c_{ma} + c_f] \times \\ &\quad P((i + \frac{m+1}{2})T < T_f < S_{(j+l-1)N+n}) \end{aligned} \right\} \quad (18)
\end{aligned}$$

The expected cycle cost of a defective renewal:

$$\begin{aligned}
E(C_r) &= \left\{ \begin{aligned} &\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^{N-1} [(i+1)c_{mi} + (j-1)c_{ma} + c_r] \times \\ &\quad P(T_r = S_{(j-1)N+n}) + \\ &\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^N \sum_{m=1}^{\infty} [(i+m+1)c_{mi} + (j-1)c_{ma} + c_r] \times \\ &\quad P(T_r = S_{(j+l-1)N+n}) + \\ &\sum_{i=1}^{\infty} \sum_{j=1}^{\infty} [ic_{mi} + jc_{ma} + c_r] P(T_r = S_{jN}) + \\ &\sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{m=1}^{\infty} [(i+m+1)c_{mi} + jc_{ma} + c_r] \times \\ &\quad P(T_r = S_{(j+l)N}) \end{aligned} \right\} \quad (19)
\end{aligned}$$

The expected cycle length of a failure renewal:

$$E(L_f) = \left\{ \begin{aligned} & \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \int_0^T [iT + u] P(iT + u < T_f < iT + u + du) + \\ & \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^N \int_0^{S_{(j-1)N+n} - (i+1)T} [(i+1)T + u] \times \\ & P((i+1)T + u < T_f < (i+1)T + u + du) + \\ & \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} \sum_{m=1}^{\infty} \int_0^{\frac{m}{2}} [(i+m)T + u] \times \\ & P((i+\frac{m}{2})T + u < T_f < (i+\frac{m}{2})T + u + du) + \\ & \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \int_0^{S_{(j+l-1)N+n} - (i+1)T} [(i+\frac{m+1}{2})T + u] \times \\ & P((i+\frac{m+1}{2})T + u < T_f < (i+\frac{m+1}{2})T + u + du) \end{aligned} \right\} \quad (20)$$

The expected cycle length of a defective renewal:

$$E(L_r) = \left\{ \begin{aligned} & \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^{N-1} S_{(j-1)N+n} P_1(T_r = S_{(j-1)N+n}) + \\ & \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{n=1}^{\infty} \sum_{m=1}^{\infty} \sum_{l=0}^{\infty} S_{(j+l-1)N+n} P_2(T_r = S_{(j+l-1)N+n}) + \\ & \sum_{i=1}^{\infty} \sum_{j=1}^{\infty} S_{jN} P_1(T_r = S_{jN}) + \\ & \sum_{j=1}^{\infty} \sum_{i=1}^{\infty} \sum_{m=1}^{\infty} \sum_{l=0}^{\infty} S_{(j+l)N} P_2(T_r = S_{(j+l)N}) \end{aligned} \right\} \quad (21)$$

The long run expected cost per unit time can be derived as

$$c(N, T) = \frac{E(C_f) + E(C_r)}{E(L_f) + E(L_r)} \quad (22)$$

which can be regarded as an objection function about decision variables T and N . The minimum long run expected cost per unit time can be solved by selecting optimal decision variables.

IV. NUMERICAL EXAMPLE

In this section, numerical examples are presented to indicate the practical purpose of this model. First, a comparison between this model and the two-level periodic inspection model is shown to demonstrate the correctness and genericity of this model. Next, an example is given to prove the maintenance action is implemented at the completion of the current job is more economical than being taken immediately

A. Verification of the Correctness and Generality of the Model

The two-level periodic inspection model (Model 1) is proposed in [14], which can be simply described as:

- The failure process is the same as this model.
- The system is subjected to minor inspections at interval T and major inspections at interval NT .
- The minor inspection can identify the minor defective stage with a limited probability γ but can always identify the severe defective stage completely. The major inspection can reveal both the minor and severe defective stages perfectly.

- If the component is found to be in the minor defective stage by an inspection, the subsequent minor inspection interval is shortened. For simplicity, the interval is shortened to be half of the current interval.
- When the component is found to be in the severe defective stage by either inspection, it is replaced immediately.

To keep consistent with Model 1, the working times of jobs in the proposed model is set to equal to the minor inspection interval, i.e. $Y_j \equiv T$, ($j = 1, 2, \dots$). Thus the major inspection is implemented at the completion of every N jobs, which means the major inspections happen at NT , that is $S_{jN} = jNT$, ($j = 1, 2, \dots$). This process and result is the same as the situation where executing jobs with random working times is not considered. In this case, (1) can be simplified by putting $Y_j = T$ and $S_{jN} = jNT$ into it:

$$P(iT < T_f < (i+1)T) = \sum_{k=0}^i P \left\{ \begin{aligned} & kT < X_1 < (k+1)T, \\ & iT - X_1 < X_2 < (i+1)T - X_1, \\ & i - k \text{ minor inspections don't detect } X_2 \text{ stage,} \\ & 0 < X_3 < (i+1)T - X_1 - X_2, \\ & (j-1)NT < X_1, \\ & X_1 + X_2 + X_3 < jNT \end{aligned} \right\} \quad (23)$$

The equation (23) is equivalent to the formula in [14], and other equations can be simplified using the same way. Then a numerical example is given, where the pdfs and parameters is the same as those in [14]. Accordingly, the three stages in failure process follow three Weibull distributions and the pdfs of the n th ($n = 1, 2, \dots$) stage is:

$$f_{X_n}(x) = \frac{\beta_n}{\alpha_n} \left(\frac{x}{\alpha_n} \right)^{\beta_n-1} e^{-(x/\alpha_n)^{\beta_n}}, \quad \alpha_n > 0, \beta_n > 0 \quad (24)$$

where α_n and β_n represent scale parameter and shape parameter, their values are listed in Table II. And other parameters are shown in Table III.

TABLE II. THE PARAMETERS OF $f_{X_n}(x)$

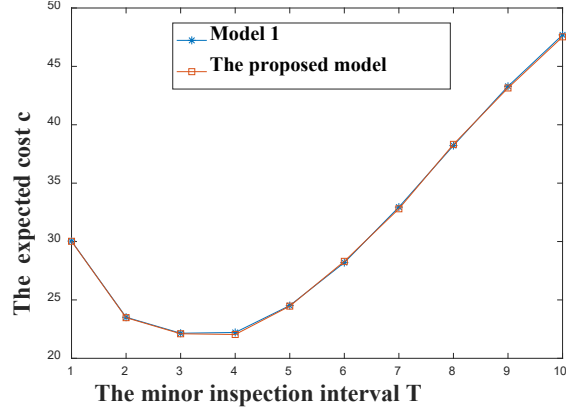
$f_{X_1}(x)$		$f_{X_2}(x)$		$f_{X_3}(x)$	
α_1	β_1	α_2	β_2	α_3	β_3
15.83	1.47	9.61	1.95	6.02	2.81

TABLE III. THE VALUES OF OTHER PARAMETERS

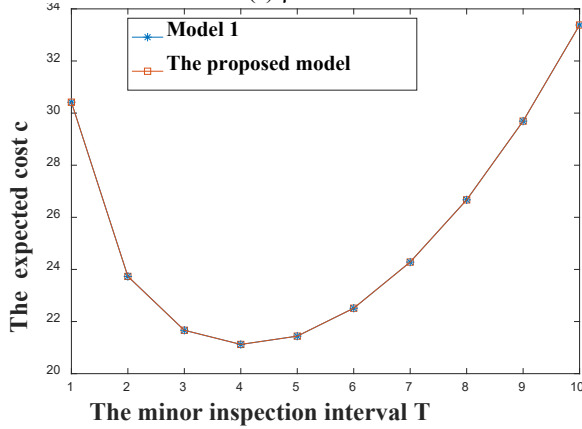
γ	N	c_{mi}	c_{ma}	c_f	c_r
0.4	5	8	20	3000	400

A simulation is conducted to verify the correctness and generality of the model presented in this paper, and the results are shown in Fig. 9(a). Except for slight differences, the expected cost per unit time of these two models are nearly identical. These slight differences result from the randomness that minor inspection identify minor defective stage with a probability $\gamma = 0.4$. If $\gamma = 0$ or $\gamma = 1$, the results of simulations are exactly equal, as shown in Fig. 9(b) and Table IV. This demonstrates that the process and result of the proposed model can be equivalent to the model where the

random working times of jobs are not considered by set the working times of jobs a suitable uniform fixed value. It proves that the model proposed in this paper is correct and more general.



(a) $\gamma = 0.4$



(b) $\gamma = 1$

Figure 9. The simulation results of Model 1 and proposed model

TABLE IV. THE EXPECTED COST PER UNIT TIME OF THESE TWO MODELS

Inspection interval	Model 1		The proposed model	
	$\gamma = 1$	$\gamma = 0$	$\gamma = 1$	$\gamma = 0$
T=1	30.3842	29.7023	30.3842	29.7023
T=2	23.6528	23.2760	23.6528	23.2760
T=3	21.5960	22.3176	21.5960	22.3176
T=4	21.1148	24.1817	21.1148	24.1817
T=5	21.4266	28.4413	21.4266	28.4413
T=6	22.4595	34.4589	22.4595	34.4589
T=7	24.3611	41.3517	24.3611	41.3517
T=8	26.7419	48.5990	26.7419	48.5990
T=9	29.5430	55.4060	29.5430	55.4060
T=10	32.8626	60.6969	32.8626	60.6969

B. The Significance of the Postponed Maintenance

In this part, an example is presented to prove that the proposed model is more economical than the model without postponed maintenance (Model 2). Model 2 means that

maintenance action is implemented immediately once the system is identified in the severe defective stage. From [7], the distribution of random times of jobs could be described as exponential distribution, i.e. $F(t) = P(Y_j \leq t) = 1 - e^{-\theta t}$. As before, the parameters is given firstly.

TABLE V. THE VALUES OF OTHER PARAMETERS

γ	N	c_{mi}	c_{ma}	c_f	c_r	θ
0.4	10	8	20	3000	400	2

As shown in the Table V, $\theta = 2$ signifies that the expected value of the random times is 0.5, so major inspection is adjusted to be implemented at the completion of every 10 jobs, i.e. $N = 10$. In this way, the expected interval of the major inspection is 5 unit times, which is in accordance with last example.

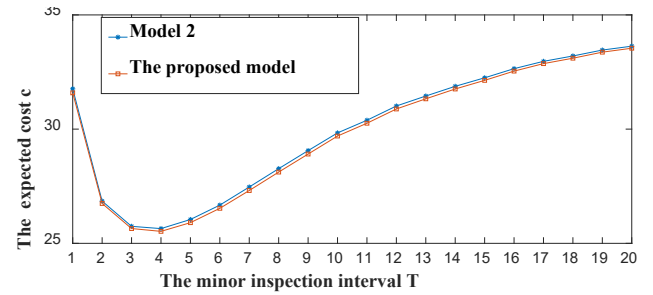


Figure 10. The simulation results of the proposed model with Model 2 for the same renewal cost ($c_r = 200$)

From the simulation results, the expected cost per unit time of the proposed model is lower. The expected length of renewal process will be longer due to postponed maintenance and its contribution outweighs the failure risk that postponed maintenance would result in. In this simulation, the renewal cost of the system being executing job is the same as that is renewed at the completion of current job. However, the cost may be lower when maintenance action happens at the completion of current job in practice. For example, the machining component will be abandoned if a replacement is arranged when machine tool is in a machining process and implementing a maintenance action is almost impossible for a plane conducting a flight mission or a train being on a journey. Therefore, when the executing job have been finished, maintenance action can be considered happens at downtime, which is more convenient as well as flexible and would lead lower cost. It is adopted in [14] that the cost of the system be renewed at planed down time is half of the cost of the maintenance be conducted immediately once the severe defective stage is identified. Accordingly, the renewal cost at the completion of current job could be half of the renewal cost when it is being executing a job. Based on this, some parameters are adjusted: c_r denotes the cost of system being renewed at the completion of current job, while c_{r2} represents the cost of the system being renewed when it is executing a job, and the values of these parameters are $c_r = 200$, $c_{r2} = 400$. Leaving the other parameters intact, another simulation result is given. As Fig. 11 shown, more economic benefit can be obtained.

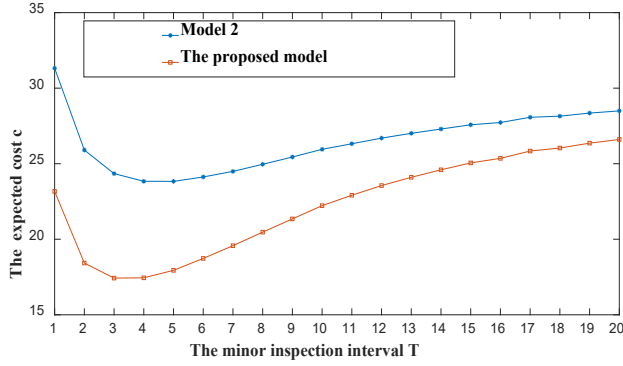


Figure 11. The simulation results of Model 2 and the proposed model for different renewal cost ($c_r = 200$, $c_{r2} = 400$).

V. CONCLUSIONS AND FURTHER RESEARCH

A new model is proposed by combining the random and periodic inspection policy based on a three-stage failure process. Accordingly, an objective function is established to minimize the long run expected cost per unit time. The optimum solution can be obtained by selecting appropriate decision variables T and N . In this model, the failure process of the system is divided into three independent stages: normal, minor and severe defective stages. The inspection with different depths and levels is introduced: the minor inspections happen at fixed interval and major inspections are implemented at the completion of every N jobs. The minor inspection is imperfect inspection, which can identify minor defective stage with a certain probability and distinguish severe defective stage perfectly. The major inspection detect the system comprehensively, which can always reveal minor and severe defective stage perfectly. If the system is found in minor defective stage, the minor inspection interval will be shortened. When inspection detect the system in severe defective stage, instead of maintaining the system immediately, the maintenance action will be implemented at the completion of the current job. Then numerical examples are given to demonstrate the correctness, generality and economy of the model proposed in this paper. The distributions and parameters used in the simulations is almost consistent with the literatures [7, 14]. Comparing with the two-level periodic inspection model, the correctness and generality of the model is verified. The other example is presented to prove that the proposed model with postponed maintenance may be more economical than the model without postponed maintenance.

The further research could be studied in the following areas. First, considering the actual production, some work could be conducted to select or establish appropriate distributions and parameters based on field data, so the model can be better applied to practical engineering problems. Second, parameters can be analyzed to decide which model to be choose under different parameter values. Because the same system with the same process should match different model under different parameter values. For example, for the model presented in this paper, if the expected length of severe

defective stage is shortened or the cost of the failed renewal is raised obviously, the failure risk or the long run expected cost per unit time will also increase remarkably. In such a situation, taking maintenance as soon as possible would be a more proper approach. Therefore, it may not make much sense to obtain the optimal solution without parameter analysis. Last, random inspection policies could have a wider application by introducing other maintenance policies to the system executing jobs with random working times.

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